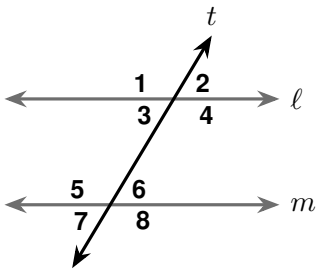
Parallel Lines & Transversals — Notes

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: When the lines are parallel*

Key Idea



If $\ell \parallel m$:

Corresponding angles are _____.

Alternate interior and alternate exterior angles are _____.

Same-side interior angles are _____ (add to _____).

☞ *Video: The easy way to see it*

Key Idea

Every angle is either _____ or _____.

Two acute angles are _____. Two obtuse angles are _____. Acute + obtuse = _____.

☞ *Video: Find the angle*

Watch Me — Worked Example

$\ell \parallel m,\ m\angle 1 = 110°$. Find $m\angle 5$ and $m\angle 6$.

$\angle 1$ and $\angle 5$ are _____ $m\angle 5 =$ _____

$\angle 5$ and $\angle 6$ are a _____ $m\angle 6 = 180 - 110 =$ _____

☞ *Video: With algebra*

Watch Me — Worked Example

Alternate interior angles: $(3x + 15)°$ and $(5x - 25)°$.

$3x + 15 = 5x - 25$ alternate interior angles are _____

$40 = 2x,\ x =$ _____

angle $= 3(20) + 15 =$ _____ answer the question they asked

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☞ *Video: Watch out*

Common Mistake

Same-side interior angles are _____, not equal. Set them to add to _____.

☞ *Video: Your turn*

Your Turn

$\ell \parallel m$. Same-side interior angles are $(2x + 10)^\circ$ and $(3x - 5)^\circ$. Find x and both angles.

$(2x + 10) + (3x - 5) =$ _____ $\Rightarrow\ 5x + 5 = 180\ \Rightarrow\ x =$ _____

Angles: _____ and _____

Notice

Remember: Parallel lines: corresponding and alternate angles are **congruent**; same-side interior angles are **supplementary**. Acute $=$ acute, obtuse $=$ obtuse, acute $+$ obtuse $= 180^\circ$.

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